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Non-inferiority, comparative effectiveness study of intravenous ketamine v. intranasal esketamine for treatment-resistant depression: The EQUIVALENCE trial protocol

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ABSTRACT

Ketamine is a powerful and rapid-acting antidepressant but has no regulatory authorization for any psychiatric illness. The S-enantiomer of ketamine, esketamine, was approved by the FDA in 2019 as adjunctive therapy for treatment-resistant depression. Substantial controversy exists as to whether there is a significant clinical difference between intravenous (IV) ketamine racemate and intranasal esketamine. The EQUIVALENCE protocol will directly compare these therapies for treatment-resistant depression (TRD). Patients with TRD are randomized (1:1) to receive 8 treatments of IV ketamine or IN esketamine over a 4-week period. The primary outcome is change in depression severity as measured by a patient-reported outcome, the Quick Inventory of Depressive Symptomatology (QIDS) after 4 weeks of treatment. The study is designed as a non-inferiority study, with the FDA-approved therapy, esketamine, considered the standard. With a sample size of 400 total (200 per group), the study is powered to declare non-inferiority within a margin of 1.6 on the QIDS. If IV ketamine is found to be non-inferior to esketamine, a subsequent statistical test will examine whether ketamine is superior. Secondary outcomes include response and remission rates, anxiety severity, quality of life, and patient satisfaction, acceptability, and tolerability. We will also examine whether TRD with anxious distress responds differentially to one treatment or the other and the role that patient preference plays in outcomes. Results of the EQUIVALENCE study will have important implications for patient choice, health insurance coverage policies, and clinical care for TRD.

1. Major depressive disorder and treatment-resistant depression

Major depressive disorder (MDD) is associated with significant morbidity and mortality [1] and is a leading cause of disability worldwide. An estimated 264 million people suffer from the disorder globablly [2]. In the United States, the total economic cost of depression in 2012 was estimated at $188 billion [3]. A significant proportion of patients diagnosed with MDD do not respond to standard early-line treatments.

The diagnosis of treatment-resistant depression (TRD) is generally given after two antidepressant treatments fail to produce an adequate response for a patient. Estimates suggest TRD comprises 12–40% of patients with depression [4,5]. Compared to patients with treatment-responsive depression, patients with TRD are at greater risk of attempting suicide [6], significantly higher risk of hospitalization [7,8], spend longer time periods in depressed states and are more likely to experience job loss [7]. Unfortunately, the therapeutic options for patients suffering with TRD remain limited. The largest antidepressant effectiveness trial to date showed that over 75% of TRD patients will not achieve remission after two additional rounds of treatment [4].

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1.1. Ketamine

A major advance in the approach to treating TRD was the discovery of ketamine as a rapid-acting antidepressant. Since 2000, a growing number of studies show that low doses of racemic ketamine delivered intravenously (IV) have rapid and robust antidepressant effects in TRD [9–13]. Based on the growing evidence of IV ketamine's antidepressant effects, providers increasingly began offering IV ketamine as a therapy for TRD in the early 2010s [14,15]. Recently, a comparative effectiveness trial showed that ketamine was non-inferior to electroconvulsive therapy in moderately severe TRD [16].

1.2. Esketamine

Janssen, LLC, initiated a large-scale clinical research program that led to FDA approval of esketamine for TRD in 2019. (Esketamine received supplemental approval for major depression with suicidal ideation in 2020 and as monotherapy in 2025.) Esketamine is the S-enantiomer of racemic ketamine and is administered intranasally. Recently completed post-approval open-label trials (covering 3777 patient-years of esketamine exposure) provide further support for the long-term efficacy and safety of esketamine [17]. Further information on the safety of esketamine from over 58,000 patients during the 5 years following FDA approval was recently published [18]. Another large randomized trial of 676 TRD patients found esketamine superior to quetiapine in remission at both 8 (primary outcome) and 32 weeks, and lower rate of treatment discontinuations [19]. This suggests esketamine has enhanced efficacy and tolerability compared to a widely used class of pharmacological treatment available for patients with TRD.

1.3. Uncertainty regarding comparative effectiveness

Despite the availability of an FDA approved therapy in esketamine, considerable controversy remains regarding whether there is a meaningful clinical difference between IV racemic ketamine (which is not FDA approved but is increasingly used) and esketamine. This growing controversy has 1) led to inconsistent messaging to patients, 2) contributed relcutance among some providers to adopt the FDA-indicated treatment and 3) created confusion among third-party payers regarding coverage policies for ketamine and esketamine treatments.

The ability to resolve this controversy is presently limited by a lack of data that would allow a direct comparison of the two treatments' effectiveness and acceptability in treating individuals with TRD. Some clinics have published real-world outcome data in non-randomized cohorts of patients receiving either intravenous ketamine or intranasal esketamine [20,21]. However, these reports may suffer from bias by unmeasured confounding. All else being equal, patients who receive IV ketamine are of higher socio-economic status than those who receive esketamine because of the much broader insurance coverage of the latter. And socio-economic status is a powerful predictor of health outcomes [22]. Hence, any differences in outcomes may be attributable to baseline differences in socio-economic status as much as differences in treatment efficacy from the drugs.

Although there are some well-founded hypotheses suggesting why one ketamine-related treatment approach could be superior to the other, the marked differences in study designs, clinical populations and the degree of scientific rigor employed in the collection of the two databases to date prohibit meaningful comparisons.

Here we describe the protocol for a comparative effectiveness, randomized, non-inferiority trial comparing intravenous ketamine to intranasal esketamine in TRD: The EQUIVALENCE study.

1.3.1. Trial overview and management

The EQUIVALENCE study is a multi-site comparative effectiveness study in patients with TRD, wherein treatment-seeking patients are randomly assigned (1:1 ratio) to receive IV ketamine or IN esketamine. Given the different treatment administration routes, treatment arms are not blinded at the clinician or patient level. The study was approved by a central IRB (Western Copernicus Global). Yale functions as the Data Coordinating Center (through the Yale Center for Analytical Sciences) and Clinical Coordinating Center (through Yale Center for Clinical Investigations [YCCI], an academic Contract Research Organization). An independent Data Safety and Monitoring Board oversees the conduct of the study. On review by regulatory authorities, it was determined that the study should be conducted under an Investigational New Drug application (#172038); during this process, the FDA required modifications from the original protocol, including a maximum lifetime and dose limit exposure to ketamine. The trial is registered on the US Clinical Trial Registry (NCT06713616).

1.4. Input from stakeholder council

A Stakeholder Council was organized, comprised of study investigators, patient and family advocates, providers, insurance company representatives, policy makers, and industry representatives. As early as the point of planning and design of the trial, input from members of the stakeholder council has been sought and incorporated into the trial. Decisions regarding the outcome assessments used, the study design (including the option to switch treatments if a patient did not respond [<50% improvement in symptom severity since baseline] or tolerate the initially assigned treatment), and strategies for recruitment have been made with critical input from the stakeholders. During the trial, the Stakeholder Council meets 3–5 times per year to discuss study progress, provide give feedback on enrollment, recruitment, and retention strategies, and discuss ways to disseminate study findings to a wide variety of audiences.

2. Recruitment and eligibility

Treatment-seeking patients at academic and community sites will be recruited for inclusion into the EQUIVALENCE study. After signing informed consent, patients will be eligible for enrollment based on the following. Key inclusion criteria include 1) age 18 or older, 2) diagnosis of major depressive disorder (diagnosed by Mini International Neuropsychiatric Interview [MINI]) refractory to two or more antidepressant trials (as judged by the MGH Antidepressant Treatment Resistance Questionnaire (ATRQ), 3) moderate or severe MDD (based on initial Montgomery-Asberg Depression Rating Scale [MADRS] of $\geq$25), 4) clinically appropriate for esketamine or ketamine, 5) adequate contraception among females of childbearing potential.

Key exclusion criteria include 1) diagnosis of bipolar disorder or psychotic disorder (i.e., schizophrenia, schizoaffective disorder), 2) active or recent (within 12 months) substance use disorder (other than nicotine), 3) uncontrolled hypertension, 4) intracerebral hemorrhage or aneurysmal vascular disease, and 5) positive urine toxicology at screening visits, except for prescribed substances.

Diagnoses other than bipolar disorder, schizophrenia/schizoaffective disorder are permitted so long as depression is the predominant diagnosis. Given that this is a real-world comparative effectiveness trial, concurrent psychotherapy and adjustments to medications are permitted.

Full inclusion / exclusion criteria can be found in the supplement.

2.1. Study design

2.1.1. Real world trial

The study has been designed to be a real-world, effectiveness trial so that its results have direct relevance to clinical practice for patients with TRD. Within the framework of the study and some limitations imposed by the FDA, clinicians have flexibility to adjust study drug dose or concomitant medications.

2.1.2. Randomization

All eligible patients are randomized (1:1) to either intranasal esketamine or intravenous ketamine treatment. Randomization is conducted centrally through a secure electronic data management system (REDCap) and is stratified by site and the presence/absence of anxious distress (as measured by the MINI). Given the differing routes of administration and nature of the treatments, treatment arms cannot be blinded at the patient or clinician level.

2.1.3. Equipoise between treatment arms

Given the open-label nature of this study, considerable effort is exerted by study staff to maintain therapeutic equipoise. No assumptions have been made for any of the two treatments being superior to the other. Also, the conditions and implementation of the two treatment arms were made as similar as possible, with treatments being delivered in the same treatment space.

2.1.4. Non-inferiority Trial

This trial is designed as a non-inferiority trial, which is the most appropriate design to test whether an intervention which may have other advantages beyond overall treatment effects (e.g., cost, tolerability) against a gold standard treatment [23,24]. For this trial, we selected the gold standard to be esketamine based on its FDA approval status. If ketamine is found to be non-inferior to esketamine, a subsequent statistical test is planned to assess whether ketamine is superior to esketamine.

2.1.5. Study treatments

During the EQUIVALENCE protocol, patients will receive up to eight treatments over four weeks at an interval of two treatments per week. This schedule is based on the FDA label for esketamine and evidence showing that providing ketamine three times per week is not superior to providing ketamine twice per week [25]. Ketamine is delivered in the same physical space and by the same providers as esketamine. Patients may remain on their concomitant psychotropic regimen throughout the treatment, with a few exceptions (noted below).

Ketamine is delivered intravenously, starting at the most commonly utilized dose of 0.5 mg/kg infusion over a 40-min period. Investigators may modify the dose if clinically warranted up to a maximum of 60 mg (a limit imposed by the FDA). Esketamine is delivered intranasally, with the starting dose of 56 mg (the starting dose used in the TRD trials that led to initial approval). Investigators may decrease (for tolerability) or increase (for lack of efficacy) the esketamine dose to a maximum of 84 mg, in line with the FDA-approved prescribing information.

For both treatments, patients are required to be present at the clinic and must be observed for 2 h following dosing. At every visit, patients are assessed by clinical providers to evaluate therapeutic response, adverse events, suicide risk, and the emergence of dissociative or psychotic symptoms.

2.1.6. Option to switch treatments

There is a potential for the investigator to switch the participant to the opposite treatment for non-response or inability to tolerate the assigned treatment following treatment with the originally assigned therapeutic. This aspect of the study was incorporated based on input of patient stakeholders to enhance the patient-centeredness of the trial. This feature will also provide estimates of response and remission probabilities among those who don't initially respond to either ketamine and esketamine, data that stakeholders from insurance companies considered valuable. The primary outcome of the study is assessed prior to patients switching treatments and, as such, data collected following treatment switching will primarily be descriptive.

2.2. Concomitant medications

At the recommendation of the FDA, patients are not permitted to be taking theophylline, aminophylline, vasopressin, or monoamine oxidase inhibitors (MAOIs). Chronic (standing regimen) opioids (including tramadol) will not be allowed during study participation. As needed opioids will be permitted, but not within 24 h of dosing. Benzodiazepines and benzodiazepine-like drugs in common use for insomnia (e.g., zolpidem) will be permitted, but not within 12 h of dosing. Prescribed stimulants will not be permitted within 12 h of dosing. Other psychotropic medications are permitted as clinically indicated.

2.3. End of treatment visit

Patients complete an End-of-Treatment (EOT) visit within 3 days of the final treatment. If treatments are not stopped early, this occurs after the 8th treatment. If treatment is discontinued early, the EOT visit occurs within 3 days of the final treatment. Following this visit, patients may continue to receive esketamine or ketamine clinically, but the EOT visit marks the end of the acute treatment phase and the end of the EQUIVALENCE study. Patients are classified as responders or non-responders at the conclusion of the EOT visit based on whether they achieve ≥50% improvement or < 50% improvement, respectively.

2.4. Follow-up OBSERVE protocol

Following the EOT visit for the EQUIVALENCE protocol, patients may continue to receive maintenance ketamine or esketamine clinically, outside the EQUIVALENCE protocol. Those patients that consent will be followed in a subsequent observational study, the OBSERVE protocol. The OBSERVE protocol follows patients for an additional 6 months following the EQUIVALENCE EOT visit, with study visits at 1-, 3-, and 6-months following completion of the acute phase of treatment. (Originally, the EQUIVALENCE and OBSERVE protocols were conceived as a single protocol but the FDA required that they be split into two separate protocols. At the FDA's request, some patients will be enrolled in the OBSERVE protocol that did not participate in the EQUIVALENCE protocol.) The OBSERVE protocol is registered under a separate IRB protocol and US Clinical Trials Registry protocol (#2000038673; clinicaltrials.gov identifier: NCT06725277). At each OBSERVE visit, clinician-reported outcomes, patient-reported outcomes, and adverse events are collected, similar to the EQUIVALENCE protocol. (See Fig. 1.) (See Table 1.)

2.5. Psychometric measurements

The Quick Inventory of Depressive Symptomatology-Self Report (QIDS-SR-16) [26] will be used as the primary outcome measure. The selection of the QIDS-SR-16 is consistent with the patient-centered priorities of the Patient-Centered Outcomes Research Institute, which funds the trial. A standardized, brief instruction video was created for the study and is shown each time a patient completes the QIDS-SR-16.

For clinician administered scales, all raters are required to complete training pertinent to each scale they would administer and document competence through a rater training program (Signant Health).

Study staff who were certified to rate the Montgomery-Asberg Depression Rating Scale (MADRS) were required to have extensive prior clinical exposure and were also required to undergo training and independent rating of patient interviews. Annual calibration exercises using audio files of MADRS interviews assessed ongoing reliability. MADRS training and calibration was provided by a senior independent consultant with extensive experience with industry trials. Additionally, real-time quality control checks are implemented ensuring concordance between comparable items from the MADRS and QIDS-SR16. Finally, participants watch a 90-s instructional video every time prior to completing the QIDS-SR16.

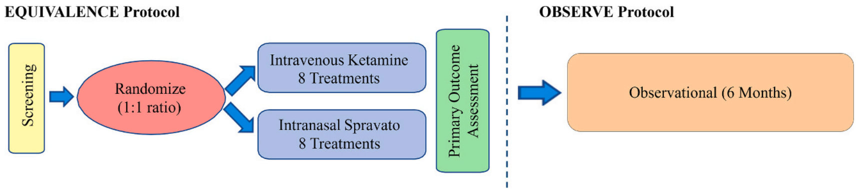

Fig. 1. Trial Schema. Following a screening period, subjects in the EQUIVALENCE protocol are randomized to 4 weeks of IV ketamine or IN Spravato. Subjects can be switched to the other treatment for non-response or poor tolerability following the primary outcome assessment (not shown in the diagram for purposes of clarity). Primary outcome will be assessed after 4-week treatment period. Following their participation in EQUIVALENCE, subjects may enter into a follow-up, observational study, the OBSERVE protocol. During this observational study, treatments are not controlled or manipulated. During the EQUIVALENCE protocol, the primary outcome (QIDS) is assessed weekly while the key secondary outcome (MADRS) is assessed every other week. During the OBSERVE protocol, the QIDS and MADRS are assessed at months 1, 3, and 6.

Table 1
Differences and similarities between intravenous ketamine and intranasal esketamine

	Ketamine (intravenous)	Esketamine (intranasal)
Scientific Evidence	Evidence mostly comprised of small studies consisting of a single dose	Evidence comprised of large Phase 2 or Phase 3 studies with multi-dose protocols trials (registered with the US FDA)
	Mostly single-site trials	Multi-site trials
	Funded by government and other non-profit organizations	Funded by pharmaceutical industry
	Most evidence with the intravenous route of Administration	Most evidence with the intranasal route of administration
	Primary timepoints evaluated were generally at 24 h after single administration	Primary timepoints evaluated were after 4 weeks of treatment
Clinical Characteristics	Requires the insertion of an intravenous catheter	Does not require intravenous catheter
	Comparatively Inexpensive (medicine only)	Comparatively Expensive (medicine only)
	Administration costs are expensive	Administration costs are expensive
	Contains the S- and R-enantiomers of ketamine.	Contains only the S-enantiomer of ketamine.
	Requires in-office administration*	Requires in-office administration (FDA requirement)
	Regulations are less stringent (no REMS)	Subject to strict safety regulations (through FDA-imposed REMS)

* While some protocols have called for at-home administration of ketamine (and some clinics provide this), this is usually oral or sublingual ketamine. We judge it is unsafe and impractical to deliver intravenous ketamine in a home setting. Intravenous remains the route of administration that has by far the largest evidence base in the published literature for ketamine and depression.

2.6. Patient-reported outcomes (PROs)

Patient-reported outcomes (PROs) utilized in the study include the QIDS-SR-16 (primary outcome), Generalized Anxiety Disorder-7-item scale (GAD-7), Patient Global Impression (PGI), Sheehan Disability Scale (SDS), Quality of Life in Depression Scale (QLDS), and Treatment Satisfaction Questionnaire for Medication, 9-item (TSQM-9). Treatment preference and expectations (using the Stanford Expectations Treatment Scale [SETS]) are assessed prior to treatment. Scales are assessed according the Schedule of Activities (Table 2 and Table 3).

2.7. Clinician-reported outcomes

Clinician-reported outcomes include the MADRS, Columbia Suicide Severity Rating Scale (CSSRS), Clinician Global Impression (CGI), Clinician Administered Dissociative State Scale (CADSS), Brief Psychiatric Rating Scale (BPRS), Modified Observer's Assessment of Alertness/Sedation Scale (MOAA/S) and the Physician Withdrawal Checklist, 20-item (PWC-20). Assessments are administered according the Schedule of Activities (Table 2 and Table 3).

2.8. Adverse events and other safety assessments

Prior to treatment, a full physical examination, including vital signs are collected. In addition, lab work, which includes complete blood count (CBC) and comprehensive metabolic panel (CMP), and urine toxicology testing are collected. Females of childbearing potential undergo urine pregnancy testing. Patients with active cardiovascular conditions undergo electrocardiogram prior to treatment. Adverse events are assessed at each study visit. Concomitant medications, and psychotherapy visits will be collected.

2.9. Statistical analysis plan and power

This trial is a non-inferiority study to determine whether ketamine is non-inferior to esketamine. The statistical hypotheses are as follows:

H0: $\Delta Q_K - \Delta Q_S \geq \delta$ (null hypothesis: ketamine is inferior to esketamine).

H1: $\Delta Q_K - \Delta Q_S < \delta$ (alternative hypothesis: ketamine is not inferior to esketamine).

Where ΔQ_K is the change from baseline in the QIDS-SR16 score (e.g., QIDS at 4 weeks minus QIDS at baseline) in the ketamine group, ΔQ_S is the change from baseline in the QIDS score in the esketamine group, and δ is the non-inferiority margin.

In this study, we will adopt a non-inferiority margin of 85% (i.e., that if ketamine retains 85% or more of the efficacy of esketamine, then this is not a clinically meaningful difference). Based on prior FDA-registered studies of esketamine [27,28], this translates to a non-inferiority margin of 1.6 QIDS points. Assuming a 2.5% significance level (one-sided) and 80% power, our projected sample size is 179 per group or 358 total. Assuming 10% attrition over the 4 weeks of the acute phase, we would need 200 patients per group, or 400 total to achieve this statistical threshold.

The primary outcome measure of depression severity improvement (measured by QIDS-SR-16) will be analyzed using a mixed-effects model with repeated measures. QIDS will be assessed weekly during the treatment period as well as EOT and 3–5 weeks after EOT. The repeated measures mixed model will include treatment, time, and time-by-group interaction as fixed effects, with a random patient effect. The model will also include baseline covariates of depression severity, sex, age, clinical site, treatment dose, anxious distress, and concomitant medications. A one-sided, 95% confidence interval for the difference in changes in QIDS at 4 weeks between IV ketamine and esketamine will be evaluated to see if it contains the inferiority boundary. If the primary hypothesis (i.e., that ketamine is non-inferior to esketamine) is supported at the pre-specified threshold, a follow-up analysis will be conducted to

Table 2
Abbreviated* schedule of activities, EQUIVALENCE protocol

	SCREENING	TREATMENT PHASE**								
	Screening	Week 1		Week 2		Week 3		Week 4		
Item		Visit 1 (Baseline)	Visit 2	Visit 3	Visit 4	Visit 5	Visit 6	Visit 7	Visit 8	EOT (Day 29)
Informed Consent	X									
QIDS-16 (Primary outcome)		X		X		X		X		X
MADRS	X	X				X				X
MINI (7.0.2)	X									
ATRQ	X									
CSSRS	X	X	X	X	X	X	X	X	X	X
Demographics and Medical History	X									
Preference of Treatment	X									
Vital signs	X	X	X	X	X	X	X	X	X	X
Urine toxicology and HCG	X	X				X				X
Concomitant Medications and Adverse Events	X	X	X	X	X	X	X	X	X	X
CGI and PGI		X				X				X
SETS		X		X						
GAD-7		X		X		X		X		X
SDS		X				X				X
TSQM-9										X
QLDS		X								X
Administration of Ketamine/Esketamine		X	X	X	X	X	X	X	X	
CADSS, MOAA/S, BPRS***		X	X	X	X	X	X	X	X	

*Full Schedule of Activities is found in the supplement. An additional follow-up visit, 3–5 weeks following EOT will also be conducted.
**If participants are switched to the opposite treatment during the Treatment Phase, the assessment schedule will be identical to the Treatment Phase for an additional 4 weeks (or until they complete the index course, whichever is shorter).
***The CADSS, MOAA/S, BPRS assessments will be administered after the infusion/insufflation for each visit where treatment is administered. The other assessments for each visit are completed before infusion/insufflation.
ATRQ – Antidepressant Treatment Response Questionnaire; BPRS – Brief Psychiatric Rating Scale; CGI – Clinician Global Impression; CADSS – Clinician Administered Dissociative State Scale; CSSRS – Columbia Suicide Severity Rating Scale; EOT – End of Treatment; GAD-7 – Generalized Anxiety Disorder, 7-item; MADRS – Montgomery Asberg Depression Rating Scale; MINI – Mini International Neuropsychiatric Interview; MOAA/S – Modified Observer's Assessment of Alertness/Sedation Scale; PGI – Patient Global Impression; QIDS-SR16 – Quick Inventory of Depressive Symptomatology, Self-Report, 16-item; QLDS – Quality of Life in Depression Scale; SDS – Sheehan Disability Scale; SETS – Stanford Expectations of Treatment Scale; TSQM-9 – Treatment Satisfaction Questionnaire for Medication (9-item).

determine if ketamine is superior to esketamine using a two-sided significance level of 0.05. The primary analysis will be conducted using the modified intention-to-treat (mITT) dataset. Any patient who has received at least one treatment with ketamine or esketamine and has at least one post-baseline depression rating will be included in mITT dataset.

2.9.1. *Secondary analyses*

As is the case with most trials of depression, it is expected that this sample will be heterogenous with respect to several key clinical and demographic characteristics. A key secondary goal of the trial will be to examine heterogeneity of treatment effect (HTE). Based on the literature, we have chosen to focus our analysis of HTE on two key factors: comorbid anxious distress and patient preference of treatment. Randomization is stratified by the presence or absence of baseline anxiety as determined by the DSM-5 MDD diagnosis specifier of anxious distress. Our analytic approach to HTE will include linear regression models, with an interaction between treatment group and the potentially relevant moderator used as a key independent variable. The dependent variable will be the change in depression severity, as measured by the primary outcome measure (QIDS-SR16) at EOT.

Additional secondary analyses will include examination of comparative efficacy as measured by change in MADRS and CGI/PGI, change in anxiety severity, response and remission rates, and specific changes in suicidal ideation and behavior. Secondary, indirect analyses of efficacy will include differences in utilization of healthcare resources and levels of disability. Continuous outcomes will be compared using mixed-effects linear regression models adjusting for the same covariates as in the primary outcome analysis. These secondary analyses will also compare the treatment on patient satisfaction, acceptability, and tolerability. In specific, adverse events of special interest (dysphoric reactions, vomiting, and elevated blood pressure beyond 180/110 or heart rate above 140) will be compared between groups.

3. Discussion

The EQUIVALENCE study will be the first, well-powered, head-to-head trial that compares the effectiveness of esketamine and ketamine for treatment-resident depression. The study is designed to provide definitive data to help settle an area of significant controversy. The primary question to be answered is whether ketamine is non-inferior to esketamine for treatment-resistant depression. If this question is answered in the affirmative, a subsequent analysis will be conducted to examine whether ketamine is superior to esketamine in this clinical context. Results of the EQUIVALENCE study will have important implications for patient choice, health insurance coverage policies, and clinical care.

Regarding the comparative effectiveness of ketamine and esketamine, existing opinion to date is fairly widespread, although rigorous data are lacking. Among the first to formally explore whether there is a meaningful clinical difference between ketamine and esketamine was conducted by Bahji and colleagues. This meta-analysis analyzed several trials of ketamine and esketamine, noting that trials using ketamine showed larger effect sizes and concluded that ketamine was more efficacious for depression. However, this study did not account for categorical distinctions between the large, multicenter, multidose, FDA-registered trials that were exclusively using esketamine and the small, single-site, single-dose non-FDA trials that were almost exclusively using ketamine [29]. Furthermore, only one study included in the meta-analysis directly compared ketamine to esketamine. This relatively small study ($N = 63$) only used a single dose (instead of the 3–4 weeks of treatment typically provided in clinical care) and found no difference between the two treatments [30]. Several clinics have analyzed their data in a retrospective, observational, non-randomized fashion [20,21].

Table 3
Abbreviated* Schedule of Events – OBSERVE Protocol

	SCREENING	FOLLOW UP PHASE		
Item	Screening	Month 1	Month 3	Month 6
Informed Consent	X			
QIDS-16	X**	X	X	X
MADRS	X**	X	X	X
CSSRS	X**	X	X	X
Concomitant Medications and Adverse Events	X**	X	X	X
CGI	X**			X
PGI	X**			X
GAD-7	X**	X	X	X
SDS	X**	X	X	X
TSQM-9		X	X	X
QLDS		X	X	X

Following participation in the EQUIVALENCE protocol, patients have the option of an additional 6 months of observational follow-up during a separate protocol, the OBSERVE protocol. This protocol does not control or manipulate treatment in any way. Originally, the EQUIVALENCE and OBSERVE protocols were conceived of as a single study, but the FDA required they be separated into two protocols. The OBSERVE protocol will recruit some patients who did not participate in the EQUIVALENCE protocol.
**Full Schedule of Activities is found in the supplement.
**The screening visit for OBSERVE coincides with the EOT visit for EQUIVALENCE; hence, the assessments marked with ** are collected once but used for both protocols.
CGI – Clinician Global Impression; CSSRS – Columbia Suicide Severity Rating Scale; GAD-7 – Generalized Anxiety Disorder, 7-item; MADRS – Montgomery Asberg Depression Rating Scale; PGI – Patient Global Impression; QIDS-SR16 – Quick Inventory of Depressive Symptomatology, Self-Report, 16-item; QLDS – Quality of Life in Depression Scale; SDS – Sheehan Disability Scale; SETS – Stanford Expectations of Treatment Scale; TSQM-9 – Treatment Satisfaction Questionnaire for Medication (9-item).

Many of these findings show a slightly greater rate of improvement with ketamine compared to esketamine. However, these reports are subject to bias from unmeasured confounding. The most likely unmeasured confounding variable that might introduce bias is socioeconomic status. Intranasal esketamine receives much broader insurance coverage than intravenous ketamine, which is generally not covered by insurance and typically requires out-of-pocket payment. As a result, patients receiving intravenous ketamine are, on average, likely to come from higher socioeconomic backgrounds compared to those receiving esketamine. Given the well-established association between socio-economic status and health outcomes, observed differences in clinical effectiveness between the two treatments may be attributable to underlying socio-economic disparities rather than to the treatments themselves [22].

In addition to its rapid-acting effects, another exciting aspect of ketamine as a therapeutic is that it appears to be efficacious in patients with MDD with anxious features. Historically, the presence of anxiety has been a negative prognostic factor in terms of therapeutic outcomes [31]. In contrast, when treated with ketamine, several studies have shown that patients with anxious depression do as well or better than patients with non-anxious depression [32,33]. Additionally, small trials show promise of ketamine as a therapeutic for generalized anxiety disorder or social anxiety disorder [34,35]. Some hypothesize that the R-enantiomer of ketamine, which is present in racemic ketamine but not esketamine, has anxiolytic effects [36]. To that end, the EQUIVALENCE trial will examine heterogeneity of treatment effect with respect to the presence or absence of anxious distress in patients with TRD and whether this effects what treatment patients and providers should select based on anxiety as a symptom.

Study participation in the EQUIVALENCE trial is restricted to individuals suffering depressive episodes who do not have bipolar disorder. Esketamine is not FDA approved as a therapy for bipolar depression, though it has been used (as well as ketamine) off-label for this clinical scenario [37,38]. The study does not formally exclude patients with current psychotic depression.

In the current study, the dose of IV ketamine is limited to 60 mg per treatment by FDA policy. In clinical use of ketamine for depression, doses that exceed this threshold are not uncommon [15]. (One report noted that some clinics report doses up to 3 mg/kg, or 210 mg for a 70 kg person [14]). Hence, the imposed dose restriction may limit the real-world nature and applications of the trial. At the same time, it is important to note that the FDA dose restriction exists because of concerns about potential for neurotoxicity with higher doses, frequencies, or cumulative exposures [39]. Given this, the current trial may be the best possible attempt to answer the question of comparative effectiveness in light of the constraints of the current regulatory framework. The OBSERVE study, which simply follows patients being treated clinically after the EQIVALENCE protocol, will provide additional information related to effectiveness and safety of higher ketamine doses and more than 8 treatments.

4. Implications of study results and conclusions

If the findings of this study indicate that IV ketamine is non-inferior to Spravato, this could influence policy makers to expand insurance coverage and hence expand treatment options for patients, including those who may benefit from or prefer IV ketamine. Expanded insurance coverage for IV ketamine may also reduce utilization of and subsequent risks associated with poorly regulated ketamine provision (e.g., take-home dosing). If findings indicate that IV ketamine is inferior to esketamine, it will help put to rest discomfort and uncertainty from patients and clinicians who believe that esketamine is not as effective as IV ketamine. Additionally, this will be powerful evidence for policy makers to sustain current coverage policies and enhance access to esketamine with greater confidence in its clinical value. Finally, the EQIVALANCE study in combination with the OBSERVE study will provide much needed information on the longer-term safety of ketamine treatments under controlled conditions.

CRediT authorship contribution statement

Samuel T. Wilkinson: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Sandhya Prashad:** Writing – review & editing, Project administration, Investigation, Conceptualization. **Rachel Dalthorp:** Writing – review & editing, Project administration, Investigation. **Lisa Harding:** Writing – review & editing, Project administration, Investigation. **Brandon Kitay:** Writing – review & editing, Project administration, Investigation. **Sagar V. Parikh:** Writing – review & editing, Project administration, Investigation. **James Dziura:** Writing – review & editing, Validation, Software, Project administration, Investigation, Formal analysis, Data curation. **Ashley Clayton:** Writing – review & editing, Project administration, Methodology, Investigation. **Geena Fram:** Writing – review & editing, Project administration, Investigation. **Valerie Lepoutre:** Writing – review & editing, Project administration, Investigation. **Julianna Brown:** Writing – review & editing, Project administration, Investigation. **Robert B. Ostroff:** Writing – review & editing, Project administration, Investigation. **Gerard Sanacora:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

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Research Institute (PCORI), its board of governors, or its methodology committee.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: (In the last 36 months, Dr. Wilkinson has received contract research funding from Oui Therapeutics for the conduct of clinical trials (administered through Yale University); he has consulted or provided DSMB services for LivaNova, Mind Medicine, Neumora, and Sooma Medical.

Dr. Harding has consulted to Janssen, Compass Pathways, and GH Reseach, Otsuka, Abbie and Lilly Pharmaceuticals.

Dr. Kitay reports consulting fees from Janssen, Biogen, Boehringer Ingelheim, and Senseye. He has received contract research funding from Otsuka Pharmaceuticals (administered through Emory University).

Dr. Prashad has served as the President of the American Society of Ketamine for Physicians, Psychotherapists and Practitioners (unpaid role).

Dr. Parikh reports clinical trial contract funding from Aifred, Janssen, Sage and Compass and consulting fees from Otsuka, Boehringer Ingelheim, and Mensante.

Dr. Sanacora has served as consultant to AbbVie, Ancora, Aptinyx, Atai, Axsome Therapeutics, Biogen, Biohaven Pharmaceuticals, Boehringer Ingelheim International GmbH, Bristol-Myers Squibb, Clexio, Denovo Biopharma, Douglas Pharmaceuticals, EMA Wellness, Embark, Engrail Therapeutics, Freedom Biosciences, GH Research, Gilgamesh, Holmusk, Intra-Cellular Therapies, Janssen, Levo therapeutics, Lilly, Merck, Navitor Pharmaceuticals, Neumora, Neurocrine, Newleos Therapeutics, Novartis, Noven Pharmaceuticals, Otsuka, Perception Neuroscience, Relmada Therapeutics, Seaport, Pharmaceuticals, Sage Pharmaceuticals, Seelos Pharmaceuticals, Supernus, Taisho Pharmaceuticals,Transcend Therapeutics, Usona Institute, Vistagen Therapeutics, and XW Labs; and received research contracts from Johnson & Johnson/Janssen, Merck, and the Usona Institute over the past 36 months. Dr. Sanacora holds equity in Biohaven Pharmaceuticals, Freedom Biosciences, Gilead, Newleos Therapeutics, Oui Therapeutics, Relmada, and Tetricus. He is a co-inventor on a US patent (#8778,979) held by Yale University and a co-inventor on US Provisional Patent Application No 047,162–7177P1 (00,754) filed on August 20, 2018, by Yale University Office of Cooperative Research. Yale University has a financial relationship with Janssen Pharmaceuticals and may receive financial benefits from this relationship. The University has put multiple measures in place to mitigate this institutional conflict of interest. Questions about the details of these measures should be directed to Yale University's Conflict of Interest office.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cct.2026.108273.

Data availability

There is no data to be shared at this time.

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